

Food Bridge

UML Diagrams: Class, Component, Deployment and Use Case

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Project Overview

Food Bridge is a web-based platform that reduces food waste by connecting people who have surplus food (restaurants, hotels, caterers, households) with NGOs that serve people in need. Volunteers handle pickup and delivery, and an administrator verifies users and monitors activity.

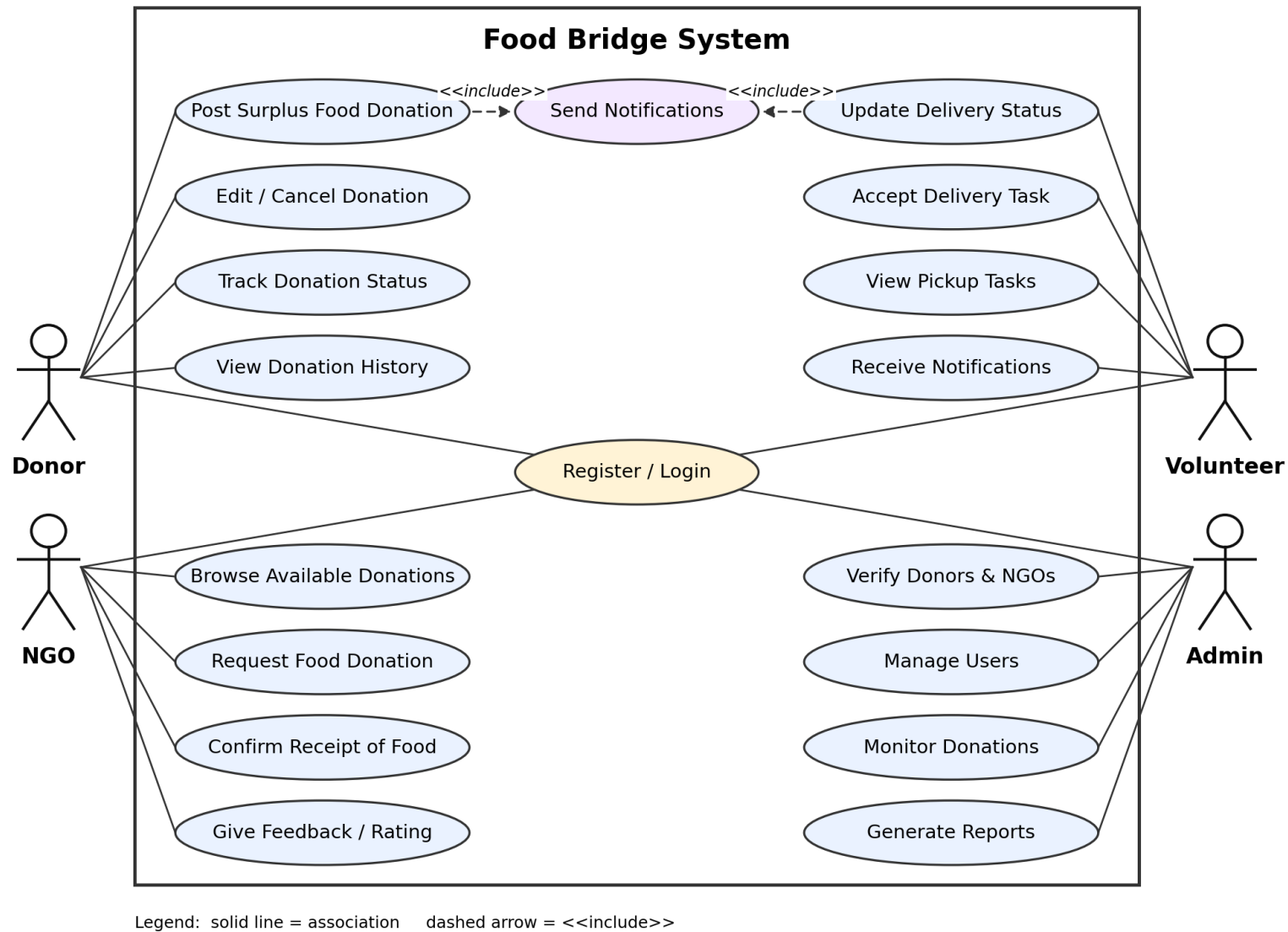
Actors

Donor posts surplus food and tracks it. **NGO** browses, requests and confirms receipt of food. **Volunteer** accepts and completes deliveries. **Admin** verifies users, monitors donations and generates reports.

Diagrams Included

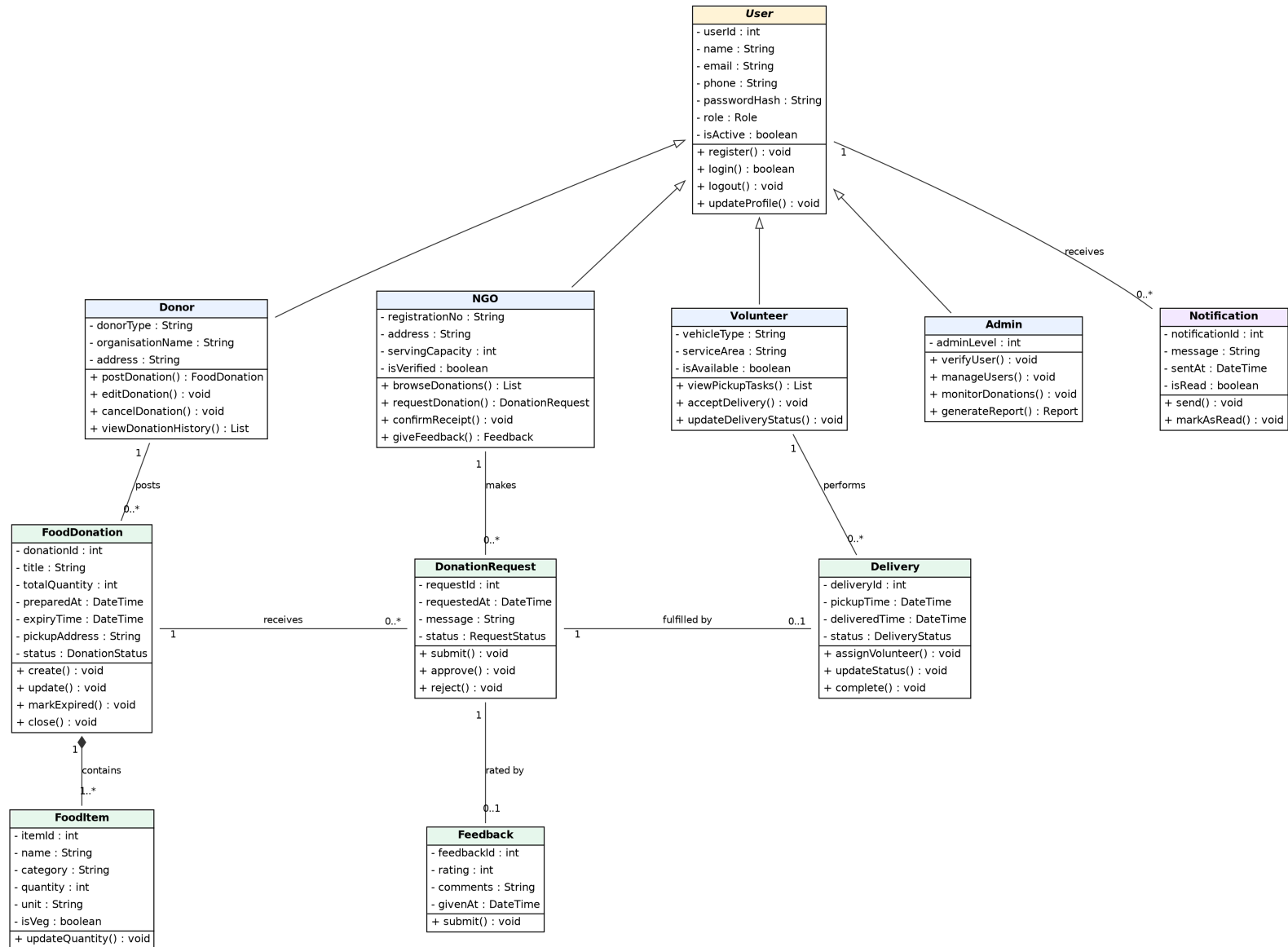
1. Use Case Diagram 2. Class Diagram 3. Component Diagram 4. Deployment Diagram

1. Use Case Diagram



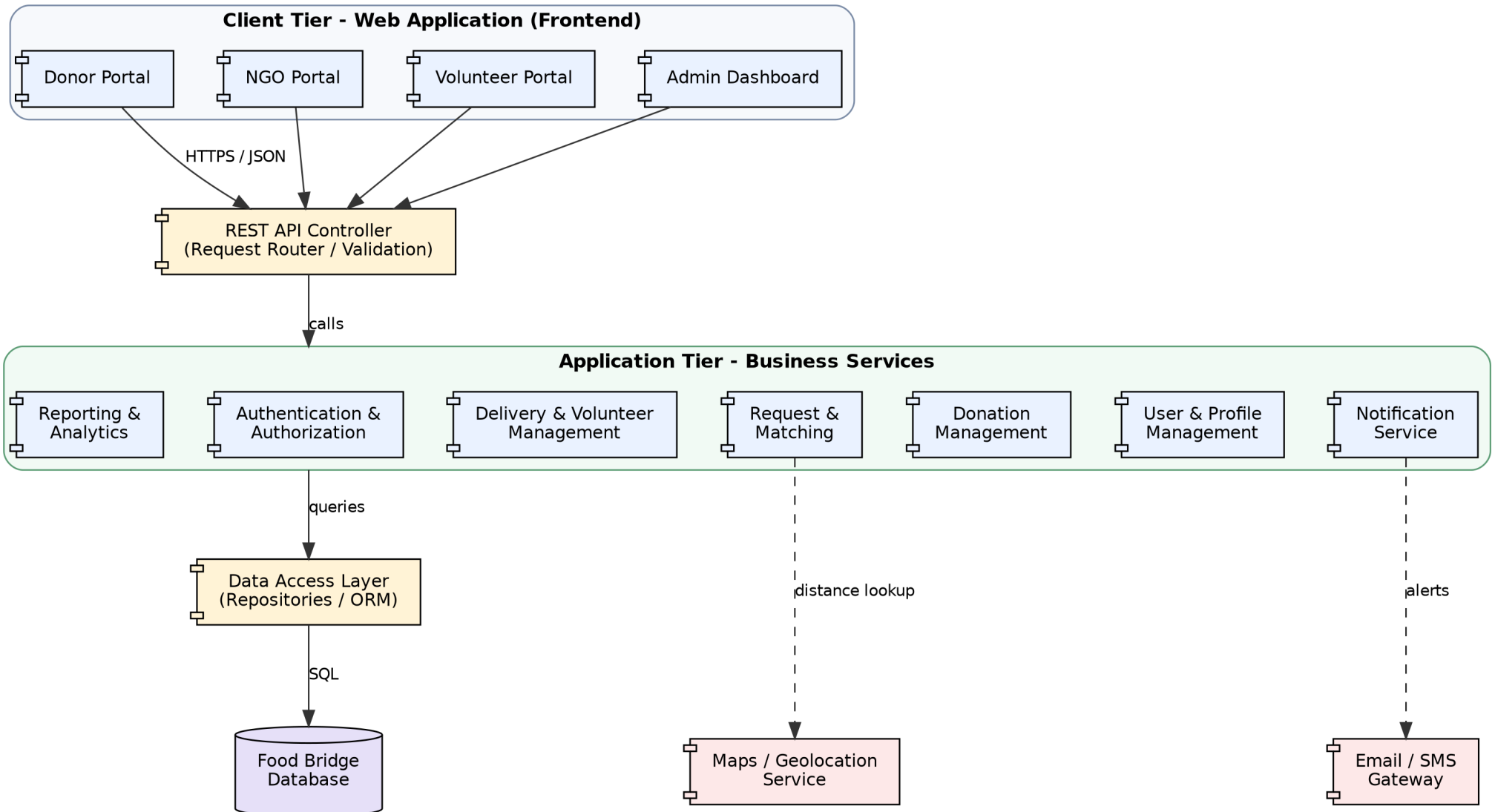
Four actors interact with the Food Bridge system. Every actor registers and logs in; posting a donation and updating a delivery both include sending notifications.

2. Class Diagram



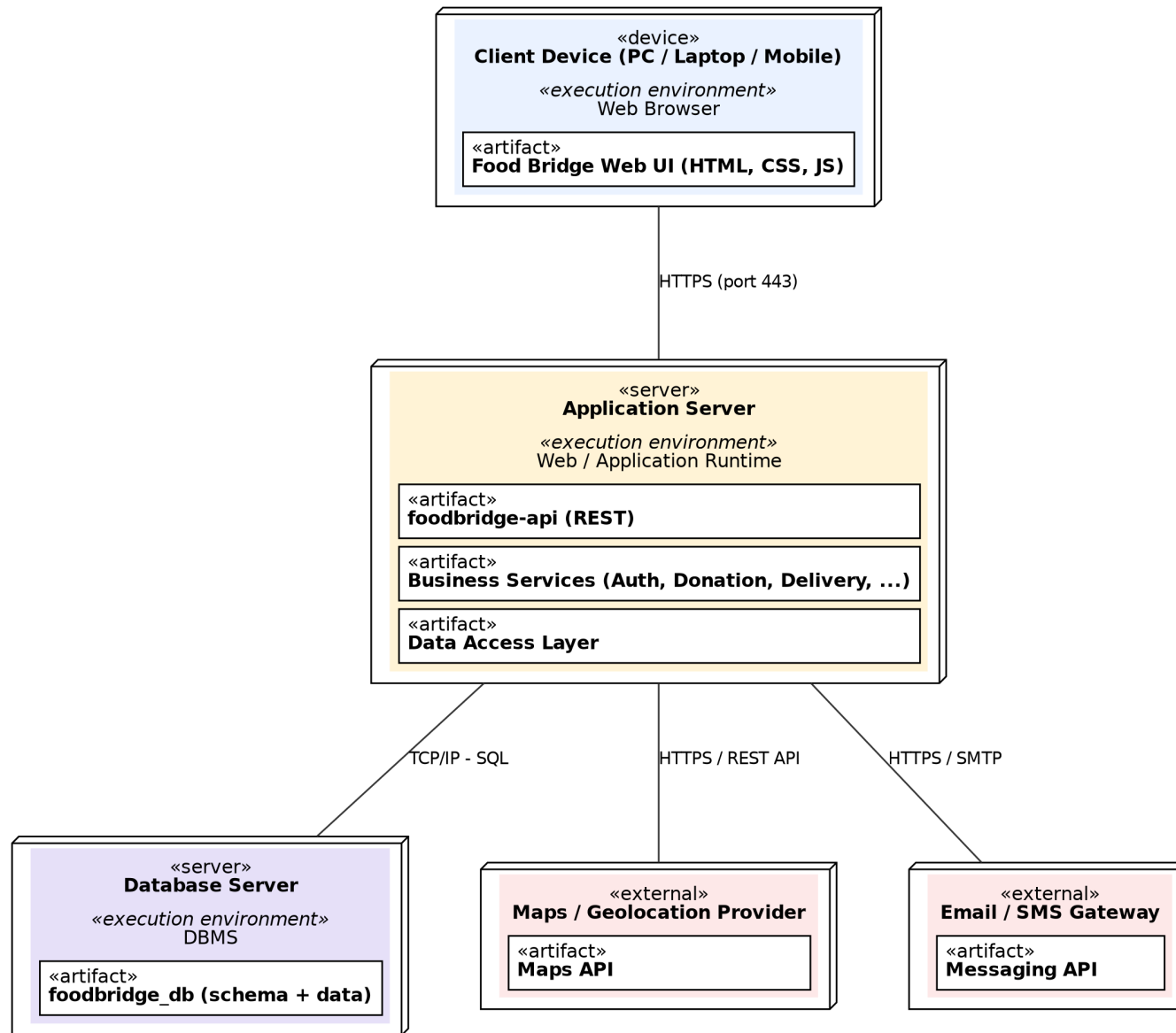
User is an abstract parent of Donor, NGO, Volunteer and Admin. A FoodDonation contains one or more FoodItems and can receive many DonationRequests; each request is fulfilled by at most one Delivery and rated by at most one Feedback.

3. Component Diagram



Role-based portals call a REST API controller, which routes to the business services. Services reach the database through a data access layer and use external Maps and Email/SMS services.

4. Deployment Diagram



Users reach the application server from a browser over HTTPS. The server talks to the database server over TCP/IP and to the external Maps and Email/SMS providers over HTTPS.